

Major Tasks in Data Preprocessing



1 Overview: Major Steps

- **Data Cleaning:** Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies.
- **Data Integration:** Combine multiple databases, data cubes, or files into a coherent store.
- **Data Reduction:** Obtain a reduced representation of the data set (Reduced volume, same analytical results).
- **Data Transformation & Discretization:** Map values to a new set of replacement values (Normalization, Concept hierarchy generation).

2 Data Cleaning

In the real world, data is often "dirty"—incomplete, noisy, or inconsistent.

2.1 Why is Data Dirty?

Dirty Data Scenarios

- **Incomplete:** Lacking attribute values (e.g., *Occupation*="").
- **Noisy:** Contains errors or outliers (e.g., *Salary*="-10").
- **Inconsistent:** Discrepancies in codes/names (e.g., *Age*="42", *Birth-day*="03/07/2010").
- **Intentional:** Disguised missing data (e.g., everyone's birthday set to Jan. 1).

2.2 Cleaning as a Process

1. **Discrepancy Detection:** Use metadata, domain knowledge, and commercial tools.
2. **Data Scrubbing:** Postal code check, spell-check, etc.
3. **Data Auditing:** Discover rules/relationships to find violators (correlation/clustering).
4. **ETL Tools:** Extraction, Transformation, and Loading through GUI or specify migrations.

3 Data Integration

Combines data from multiple sources. Key challenges include:

- **Schema Integration:** Mapping $A.cust-id \equiv B.cust-\#$.
- **Entity Identification:** Identifying real-world entities (e.g., *Bill Clinton* = *William Clinton*).
- **Resolving Conflicts:** Handling different representations or scales (e.g., *Metric* vs *British* units).

4 Data Reduction Strategies

Data Reduction

Reducing data volume while producing the same (or nearly the same) analytical results. Essential for handling terabytes of data efficiently.

- **Dimensionality Reduction:** Wavelet transforms, PCA, Feature selection.
- **Numerosity Reduction:** Regression models, Histograms, Clustering, Sampling.
- **Data Compression:** Lossy or lossless compression.
- **Data Cube Aggregation:** Summarizing data into higher levels.

5 Data Transformation

Mapping values to a new set of replacement values.

Common Methods

- **Smoothing:** Removing noise.
- **Attribute Construction:** Creating new features from existing ones.
- **Normalization:** Scaling values to a specified range (Min-Max, Z-score).
- **Discretization:** Dividing continuous attributes into intervals.

6 Discretization

Dividing the range of a continuous attribute into intervals to reduce size and prepare for analysis.

- **Attribute Types:** Nominal (un-ordered), Ordinal (ordered), Numeric (real numbers).
- **Strategies:** Supervised vs Unsupervised, Top-down (split) vs Bottom-up (merge).

Summary: Data preprocessing is a prerequisite for high-quality data mining.